

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Information and Ideas

05

10 answers · Rationale-rich · Teach from doc

Q1**D**

- D. a male tanager's appearance may function as a dishonest signal of the individual's overall fitness.

Choice D is the best answer because it most logically completes the text's discussion of Shultz's finding about male tanagers. The text explains that because carotenoids both contribute to deeply saturated feathers and offer health benefits, having deeply saturated feathers is usually "an honest signal" (a true indication) that a bird is generally fit. However, Shultz and others have found that certain male tanagers can appear to have deeply saturated feathers even if they haven't consumed a diet rich in carotenoids, thanks to microstructures in their feathers that manipulate light. If those birds aren't necessarily eating carotenoid-rich diets, they may actually be less fit than other birds that appear to have similarly saturated feathers; this suggests that a male tanager's appearance may function as a dishonest signal, or a false indication, of the bird's overall fitness.

Choice A is incorrect because Shultz's finding suggests that some tanagers can signal fitness without consuming the carotenoids that contribute to fitness, thereby making those signals dishonest, not that tanagers can give honest signals of their fitness without consuming carotenoids. Choice B is incorrect because Shultz's finding suggests that the microstructures in certain tanagers' feathers can give a dishonest signal of fitness, not that the microstructures are less effective than actual pigmentation for signaling fitness. Whether the signal of fitness is honest or dishonest has no bearing on how effective the signal is: a signal is effective if potential mates behave as though it's true, regardless of whether it's actually true. Since there's no information in the text about how potential mates respond to the dishonest signals of some tanagers, there's no support for the idea that the dishonest signals are less effective than the honest signals. Choice C is incorrect because Shultz's finding suggests that certain male tanagers may appear to be fitter than they actually are, not that scientists haven't determined why tanagers prefer mates with colorful appearances.

Q2

A

- A. Of the three films for which Carlos is credited, *Tron* features the most original music from her.
-

Choice A is the best answer because it best states the main idea of the text. The text indicates that Wendy Carlos scored three feature films: *A Clockwork Orange*, *The Shining*, and *Tron*. It also indicates that Carlos's work on *A Clockwork Orange* consisted primarily of electronic arrangements of Beethoven compositions and that very little of what she and Rachel Elkind composed for *The Shining* was used in the film. But the soundtrack for *Tron* consists largely of music composed by Carlos, so it contains more of her original music than do the soundtracks for the other two films. Thus, the main idea of the text is that of the three films for which Carlos is credited, *Tron* features the most original music from her.

Choice B is incorrect because it directly contradicts the text's claim that "very little of what she and Rachel Elkind composed for *The Shining* was used in the film." Choice C is incorrect because the text doesn't address the perceived quality of different composers, instead, it discusses how much of Carlos's original work was used in each of the three films she scored. Choice D is incorrect because it doesn't relate to any of the major themes of the text: the work Wendy Carlos did for three feature films, how much of that work was composed by Carlos, and how much of her original work was ultimately used in the film.

Q3

C

- C. “I know my own passion, it escapes me not; but, seeing that the causes are so dire, will never curb these frenzied complaints, while life is in me.”
-

Choice C is the best answer. Electra states that she “knows her own passion,” which shows that she’s aware of the intensity of her grief. But she also claims that the “causes are so dire”—meaning the reasons for her grief are so awful—that she can’t let it go, which shows that she believes her grief is justified.

Choice A is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of its intensity, but it doesn’t suggest that she believes she has a legitimate reason for feeling that way. Choice B is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of its intensity, but it doesn’t suggest that she believes she has a legitimate reason for feeling that way. Choice D is incorrect. This quotation doesn’t show that Electra believes her grief is justified. It shows that Electra is aware of the intensity of her grief, but it doesn’t suggest that she has a legitimate reason for feeling that way.

Q4

A

- A. prolonged deep sleep is likely advantageous in ways that have yet to be discovered.
-

Choice A is the best answer. It most logically completes the text. The text says that some scientists can’t explain why prolonged deep sleep is so widespread, given that the known benefits of sleep don’t seem to make up for how vulnerable it leaves an animal. This suggests that prolonged deep sleep probably has unknown evolutionary benefits that make up for the vulnerability.

Choice B is incorrect. It doesn’t logically complete the text. The text only discusses the benefits and risks of one trait: sleep. So there’s no basis for an inference about “most traits.” Choice C is incorrect. It doesn’t logically complete the text. The text says that it is already known that prolonged deep sleep is widespread in the animal kingdom. Rather, what some scientists can’t explain is WHY prolonged deep sleep is so widespread, given that the known benefits of sleep don’t seem to make up for how vulnerable it leaves an animal. Choice D is incorrect. It doesn’t logically complete the text. The text only discusses the benefits and risks of one trait: sleep. So there’s no basis for an inference about “many traits.”

Q5

D

D. They have aspirations that cannot be fulfilled because of certain limitations.

Choice D is the best answer. The text metaphorically likens humans to animals in a zoo, suggesting that humans have dreams that they cannot fulfill because they are trapped.

Choice A is incorrect. The speaker says that the lion has an “untamed heart,” but the speaker doesn’t actually mention anything about humans taming wild animals or a lack of compassion. Choice B is incorrect. The speaker doesn’t suggest that humans lack motivation. Rather, the speaker thinks that humans are “trapped” and prevented from achieving their dreams. Choice C is incorrect. The speaker doesn’t mention anything about humans becoming frustrated or lacking self-control. Rather, the speaker thinks that humans are “trapped” and prevented from achieving their dreams.

Q6

A

A. Between 1970 and 2000, the median age of first marriage rose more sharply for men in England and Wales than it did for men in the United States.

Choice A is the best answer. The student concluded that country-specific factors influence changes to median age of first marriage. This choice describes a time period when the rate of change differed between countries, suggesting that country-specific factors may have played a role in these changes.

Choice B is incorrect. This choice misreads the graph. The median age of first marriage was consistently higher for men than for women in the United States during the time period depicted. Choice C is incorrect. The student concluded that country-specific factors influence changes to median age of first marriage. However, this choice doesn’t provide any contrasts between countries and thus doesn’t support the idea of country-specific factors influencing median age of first marriage. Choice D is incorrect. The students’ conclusion is about changes that occurred during the 20th century. This choice provides broad information about the century as a whole, so it doesn’t give insight into how median age at first marriage changed over time.

Q7

B

- B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs' location.
-

Choice B is the best answer. The text says that crab burrowing in the pannes enables plants to grow there again. It also says that crab burrowing at the edges of the marsh speeds up marsh loss. This suggests that burrowing crabs can either help or hurt marshes, depending on where they're located.

Choice A is incorrect. This inference isn't supported. The text never discusses pannes of different sizes. Choice C is incorrect. This inference isn't supported. The text never suggests that crabs do more burrowing in the pannes (the areas with standing water) than they do at the edges. Rather, the text says that the burrowing that happens in the pannes is beneficial, while the burrowing that happens at the edges is harmful. Choice D is incorrect. This inference isn't supported. The text never discusses the intensity of crab burrowing, nor does it discuss the size of the panne relative to the size of the marsh.

Q8

A

- A. Continued research and development in artificial-leaf technology is needed before the devices can be widely used as an energy source.
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Choice A is the best answer because it most accurately states the main idea of the text.

According to the text, artificial leaves are a renewable energy technology that's in development and generating interest. The text goes on to state that artificial-leaf technology "is not yet commercially viable as a large-scale energy source" and that scientists from several fields are doing additional research to refine the technology. Thus, the main idea is that continued research and development in artificial-leaf technology is needed before the devices can be widely used as an energy source.

Choice B is incorrect. While the text does state that scientists from many fields are researching artificial leaves, they're doing so to address the fact that the leaves are not yet commercially viable, rather than in response to an increase in commercial use. Choice C is incorrect. Although the text does explain that artificial leaves help split water molecules into oxygen and hydrogen gas using catalysts, it doesn't argue that this process is more efficient than photosynthesis. This is merely a detail of the text and not the main idea, which is about the additional research needed to improve the cost and efficiency of these artificial leaves. Choice D is incorrect. Although the text indicates that artificial leaves mimic photosynthesis in plants, this is an incidental detail in support of the main idea that additional research is needed to make artificial leaves commercially viable.

Q9

A

- A. at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that shells were easier to collect from the beach than to harvest from the seafloor.
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Choice A is the best answer because it most effectively uses data from the table to support the researchers' conclusion about the harvesting of clamshells by Neanderthals for use as tools. The text explains that Neanderthals used clamshells to make tools and that the sturdiest, and therefore the most desirable, shells for this purpose are found on the seafloor, not on the beach. However, the researchers also concluded that the clamshell tools made from shells from the seafloor are rarer than those made from shells from the beach. Meanwhile the table shows that at each depth, the number of tools made from shells from the beach exceeds the number made from the more desirable shells from the seafloor. The fact that the more desirable shells are less common suggests that it was significantly more difficult to harvest shells from the seafloor than from the beach.

Choice B is incorrect because knowing which depth represents the period of time with the highest Neanderthal population does not help answer the question of why the Neanderthals consistently made more tools from the less desirable shells from the beach than they made from the more desirable shells from the seafloor. Choice C is incorrect because it claims that the beach shells are more durable than the seafloor shells, which contradicts the text's description of shells from the seafloor as smoother and sturdier than shells from the beach. Choice D is incorrect because knowing which depth has the most artifacts or whether the clam population fluctuated does not help explain why tools made from the less desirable shells from the beach outnumber tools made from the more desirable shells from the seafloor.

Q10

A

- A. those farmers learned the techniques from other people who were more directly influenced by Haudenosaunee practices.

Choice A is the best answer because it most logically completes the text's discussion of Euro-American farmers' use of Haudenosaunee agricultural techniques. According to the text, some Euro-American farmers were using these techniques in the early nineteenth century despite few of the farmers having seen Haudenosaunee farms. One explanation for these facts might be that the farmers developed techniques on their own that already had been invented centuries earlier by the Haudenosaunee people, but the text explicitly bars, or rules out, this explanation. If Euro-American farmers didn't learn these techniques from direct observation of Haudenosaunee practices and didn't invent the techniques independently, then the most logical explanation is that they learned the techniques from other people who were more directly influenced by Haudenosaunee practices than the farmers themselves were. Once they learned about Haudenosaunee agricultural practices, Euro-American farmers could then apply those practices to their own farming.

Choice B is incorrect because the fact that some Euro-American farmers in the northeastern United States were using Haudenosaunee techniques suggests that the techniques were likely useful for the crops the farmers raised, not that the crops typically cultivated by the farmers were not well suited to Haudenosaunee farming techniques. If the farmers' crops were ill suited to the techniques, it's unlikely that the farmers would have used those techniques. Choice C is incorrect because the text indicates only that Haudenosaunee agricultural techniques were used by Euro-American farmers in the northeastern United States, not that these techniques were widely used outside this region. Choice D is incorrect because the text states that some Euro-American farmers were using Haudenosaunee farming techniques early in the nineteenth century. This suggests that some Euro-American farmers were beginning to recognize the benefits of these techniques near the start of the century, not that such farmers only began to recognize the benefits of the techniques much later.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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